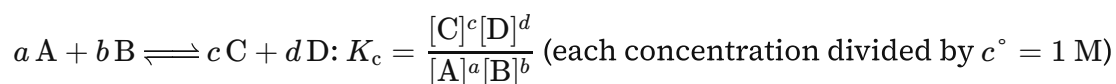


# Chemical Equilibrium and K<sub>c</sub>

K<sub>c</sub> expressions, Q versus K and ICE tables

Name \_\_\_\_\_ Class \_\_\_\_\_ Date \_\_\_\_\_

## USEFUL RELATIONSHIPS



Leave out pure solids and liquids · Reverse:  $1/K$  · Multiply coefficients by  $n$ :  $K^n$

$Q < K$ : forward ·  $Q > K$ : reverse ·  $Q = K$ : at equilibrium

Write concentrations with units (M) and give K<sub>c</sub> to three significant figures.

## A. Understanding

1. Explain what is meant by a dynamic equilibrium. Are the concentrations of reactants and products equal at equilibrium?

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## B. Writing K<sub>c</sub>

2. Write the  $K_c$  expression for: (a)  $N_2(g) + 3 H_2(g) \rightleftharpoons 2 NH_3(g)$  (b)  $2 SO_2(g) + O_2(g) \rightleftharpoons 2 SO_3(g)$  (c)  $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$  (d)  $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

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### C. Calculating $K_c$

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3. An equilibrium mixture of  $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$  contains 0.0400 M  $\text{SO}_2$ , 0.0200 M  $\text{O}_2$  and 0.160 M  $\text{SO}_3$ . Calculate  $K_c$ .

4. Using your answer to question 3, find  $K_c$  for (a)  $2\text{SO}_3 \rightleftharpoons 2\text{SO}_2 + \text{O}_2$  and (b)  $\text{SO}_2 + \frac{1}{2}\text{O}_2 \rightleftharpoons \text{SO}_3$ .

### D. Q versus K

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5. For  $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g})$ ,  $K_c = 0.0200$ . A mixture contains 0.100 M  $\text{N}_2\text{O}_4$  and 0.0200 M  $\text{NO}_2$ . Calculate Q and predict the direction of change.

### E. ICE tables

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6. 0.200 M  $\text{H}_2$  and 0.200 M  $\text{I}_2$  are mixed:  $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ ,  $K_c = 49.0$ . Find the equilibrium concentrations of all three gases.

7. 0.500 M  $\text{PCl}_5$  is placed in a container:  $\text{PCl}_5(\text{g}) \rightleftharpoons \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ . At equilibrium,  $[\text{Cl}_2] = 0.100 \text{ M}$ . Calculate  $K_c$ .

## F. Challenge

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8. One reaction has  $K = 1 \times 10^{10}$  and another  $K = 1 \times 10^{-10}$ . Describe the equilibrium mixture in each case.

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9. Using  $K_c$  from question 7, find the equilibrium concentrations when 0.300 M  $\text{PCl}_5$  is placed in a container. (You will need to solve a quadratic equation.)